

CO 2 – ASSESSMENT 3

APPLICATION – BASED QUESTIONS

1.How does cloud federation improve resource sharing and availability across datacenters?

Answer:

Cloud federation is a mechanism that connects multiple cloud providers or datacenters to share resources and services. It improves resource utilization by allowing workloads to move between different cloud environments based on demand.

Benefits of Cloud Federation:

- **Resource Sharing:** Multiple datacenters can share computing, storage, and network resources efficiently.
- **Improved Availability:** If one datacenter fails, workloads can be transferred to another datacenter, ensuring continuous service availability.
- **Load Balancing:** Workloads are distributed across different locations to avoid resource overload.
- **Scalability:** Organizations can access additional resources from federated clouds during high-demand periods.

- **Disaster Recovery:** Data and applications can be replicated across multiple datacenters for better protection.

Conclusion:

Cloud federation provides flexible resource management, high availability, and better utilization of cloud infrastructure across multiple datacenters.

2. What are the key security issues in virtualization, and how can they be mitigated?

Answer:

Virtualization allows multiple virtual machines (VMs) to run on a single physical server. However, it introduces security challenges that must be controlled.

Key Security Issues:

- **VM Escape:** An attacker may break out of a virtual machine and access the host system.
- **Mitigation:** Keep hypervisors updated and apply security patches regularly.
- **Data Leakage:** Sensitive data from one VM may be accessed by another VM.
- **Mitigation:** Use strong VM isolation and encryption techniques.

- **Hypervisor Attacks:** Attacks on the hypervisor can affect all virtual machines.
- **Mitigation:** Use secure hypervisors with proper access controls.
- **Unauthorized Access:** Unauthorized users may gain control of virtual resources.
- **Mitigation:** Implement authentication and role-based access control.

Conclusion:

Virtualization security can be improved through VM isolation, encryption, regular updates, and secure hypervisor management.

3. How does a software-defined datacenter enhance scalability and management?

Answer:

A Software-Defined Datacenter (SDDC) uses software-based control to manage computing, storage, and networking resources instead of relying only on physical hardware.

Advantages of SDDC:

Scalability:

- Resources can be added or removed dynamically based on workload requirements.
- Supports automatic scaling of applications.

Centralized Management:

- Provides a single platform to manage servers, networks, and storage.
- Reduces complexity in datacenter operations.

Automation:

- Automates resource allocation, monitoring, and deployment processes.
- Improves operational efficiency.

Flexibility:

Software-based infrastructure allows quick configuration changes without major hardware modifications.

Conclusion:

SDDC improves cloud scalability, simplifies management, and enables efficient use of datacenter resources through automation and virtualization.

4. How can identity and access control ensure data privacy in cloud-based healthcare systems?

Answer:

Healthcare systems store sensitive patient information in the cloud. Identity and Access Control (IAM) ensures that only authorized users can access medical data.

Methods to Ensure Data Privacy:

Authentication:

Uses passwords, biometrics, and multi-factor authentication (MFA) to verify user identity.

Role-Based Access Control (RBAC):

- Provides access based on user roles.
- For example, doctors can access patient records while administrative staff have limited access.

Data Encryption:

Encrypts patient information during storage and transmission.

Least Privilege Access:

Users receive only the minimum permissions required for their work.

Audit and Monitoring:

Tracks user activities and detects unauthorized access attempts.

Conclusion:

IAM protects healthcare data by controlling user access, preventing unauthorized usage, and ensuring compliance with privacy regulations.

5. How do virtualization and hypervisors support scalability and resource efficiency in cloud applications?

Answer:

Virtualization allows multiple virtual machines to run on a single physical server, while hypervisors manage and allocate resources between these virtual machines.

Role of Virtualization:

Resource Utilization:

- Multiple applications can share hardware resources efficiently.
- Reduces hardware costs and improves utilization.

Scalability:

- New virtual machines can be created quickly based on application requirements.
- Resources such as CPU, memory, and storage can be increased dynamically.

Isolation:

Each VM operates independently, improving security and reliability.

Easy Migration:

Virtual machines can be moved between servers without stopping applications.

Role of Hypervisors:

Resource Management:

Allocates CPU, memory, and storage resources to virtual machines.

VM Monitoring:

Tracks performance and manages workloads efficiently.

Load Balancing:

Distributes workloads across servers to maintain performance.

Conclusion:

Virtualization and hypervisors provide efficient resource management, flexible scaling, and improved performance for cloud applications.